

Section 3.5 Implicit Differentiation

$$\frac{d}{dx} (x^4) = 4 \cdot x^3 \quad \text{Power Rule}$$

$$\frac{d}{dx} (7x+5)^4 = 4(7x+5)^3 \cdot 7 \quad \text{Chain Rule}$$

$$\frac{d}{dx} (u^4) = 4 \cdot u^3 \cdot u' \quad \text{Implicit Differentiation}$$

Ex Find $\frac{dy}{dx}$

$$\textcircled{1} \quad x^2 + y^2 = 4 \quad \text{Take derivative of both sides.}$$

$$2x + 2y \cdot y' = 0 \quad \text{now solve for } y'$$

$$2y \cdot y' = -2x$$

$$y' = \frac{-2x}{2y}$$

$$y' = -\frac{x}{y}$$

$$\textcircled{2} \quad y^3 + e^y + \sin x = 4x$$

Differentiate both sides.

$$3y^2 \cdot y' + e^y \cdot y' + \cos x = 4$$

Put all terms with y' on one side and all other terms on the other side of the equation.

$$3y^2 \cdot y' + e^y \cdot y' = 4 - \cos x$$

Factor out y' .

$$y' \cdot (3y^2 + e^y) = 4 - \cos x$$

Divide both sides to solve for y' :

$$y' = \frac{4 - \cos x}{3y^2 + e^y}$$

③ Find $\frac{dy}{dx}$

$$\frac{y}{x} + x^2 = e^{xy}$$

$$\frac{y' \cdot x - y}{x^2} + 2x = e^{xy} (y + xy')$$

multiply both sides by x^2

$$y' \cdot x - y + 2x^3 = x^2 \cdot e^{xy} (y + xy')$$

$$y' \cdot x - y + 2x^3 = x^2 y \cdot e^{xy} + x^3 y' \cdot e^{xy}$$

Put all y' terms on the left side.

$$y' \cdot x - x^3 \cdot y' \cdot e^{xy} = x^2 y e^{xy} + y - 2x^3$$

Factor out y' .

$$y' \cdot (x - x^3 \cdot e^{xy}) = x^2 y e^{xy} + y - 2x^3$$

Divide to solve for y' .

$$y' = \frac{x^2 y e^{xy} + y - 2x^3}{x - x^3 e^{xy}}$$

What is the derivative of $\ln x$?

$$\text{Let } y = \ln x$$

$$\text{or, } e^y = x$$

Take derivative on each side.

$$e^y \cdot y' = 1$$

$$y' = \frac{1}{e^y}$$

Replace e^y with x

$$y' = \frac{1}{x}$$

$$\underline{\text{FACT}} \quad \frac{d}{dx} (\ln x) = \frac{1}{x}$$

Ex ④ Find $\frac{dy}{dx}$

$$\ln(x^2) + \sin y = \ln(\sqrt{x})$$

Take derivative on each side.

$$\frac{1}{x^2} \cdot (2x) + \cos y \cdot y' = \frac{1}{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}$$

$$\frac{2}{x} + \cos y \cdot y' = \frac{1}{2x}$$

$$\cos y \cdot y' = \frac{1}{2x} - \frac{2 \cdot 2}{x \cdot 2}$$

$$\cos y \cdot y' = \frac{-3}{2x}$$

$$y' = \frac{-3}{2x \cdot \cos y}$$

Ex ⑤ $y = 7^x$ find $\frac{dy}{dx}$

Take ln of both sides.

$$\ln y = \ln(7^x)$$

use logarithm
property to
move the exponent.

$$\ln y = x \cdot \ln(7)$$

Now take derivative on each side.

$$\frac{1}{y} \cdot y' = 1 \cdot \ln(7)$$

$$y' = y \cdot \ln(7)$$

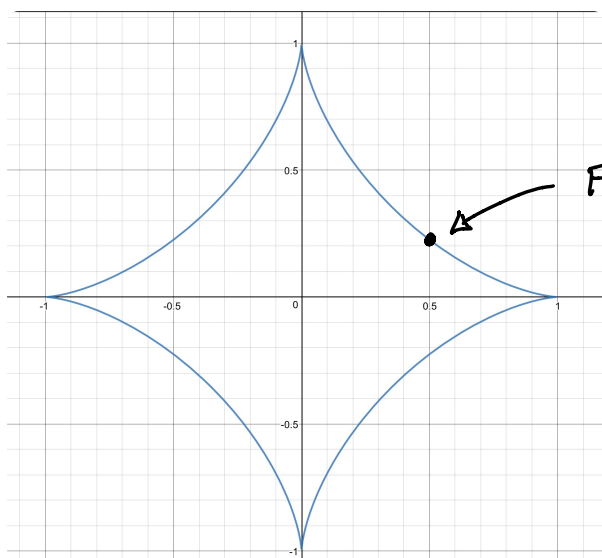
Last, replace y with 7^x .

$$y' = 7^x \cdot \ln(7)$$

Ex ⑥ Given the function

$$x^{2/3} + y^{2/3} = 1, \text{ find } \frac{dy}{dx}$$

$$\text{where } x = \frac{1}{2}.$$

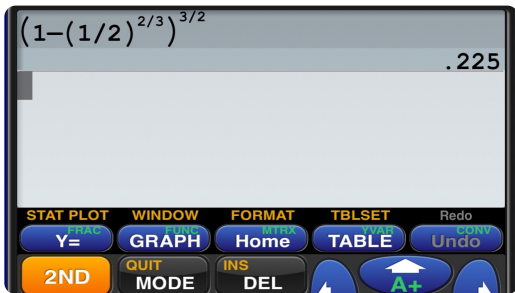


First find the y-value
at this point.

$$\left(\frac{1}{2}\right)^{2/3} + y^{2/3} = 1$$

$$y^{2/3} = 1 - \left(\frac{1}{2}\right)^{2/3}$$

$$y = \left(1 - \left(\frac{1}{2}\right)^{2/3}\right)^{3/2}$$



$$y = 0.225$$

So, the point is
 $(0.5, 0.225)$

Now find the derivative of

$$x^{2/3} + y^{2/3} = 1$$

$$\frac{2}{3} x^{-1/3} + \frac{2}{3} y^{-1/3} \cdot y' = 0$$

$$\frac{2 y'}{3 y^{1/3}} = - \frac{2}{3 x^{1/3}}$$

$$y' = \frac{y^{1/3}}{x^{1/3}}$$

Now substitute $x = 0.5$ & $y = 0.225$

$$y' = \frac{(.225)^{1/3}}{(.5)^{1/3}} = \boxed{0.587}$$

Inverse Trig Functions

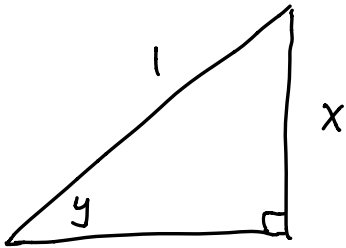
What is the derivative of

$$y = \sin^{-1} x ?$$

Take sine of both sides.

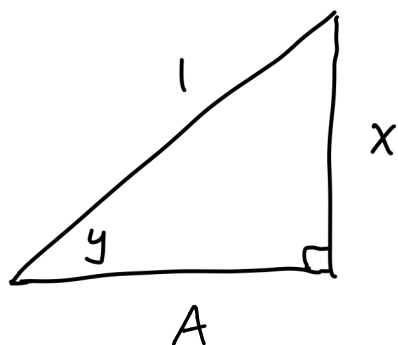
$$\sin y = x$$

This means that y is an angle.



$$\sin y = \frac{x}{1} \frac{\text{opp}}{\text{hyp}}$$

Use Pythagorean Theorem to
solve for the missing side.

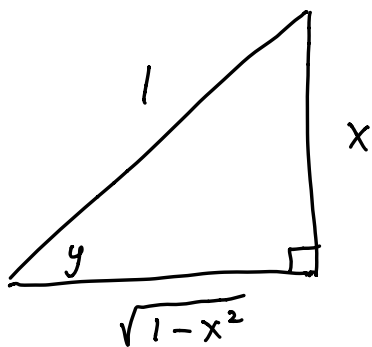


$$A^2 + x^2 = 1^2$$

$$A^2 = 1 - x^2$$

$$A = \sqrt{1 - x^2}$$

The triangle now looks like this:



$$\cos y = \frac{\sqrt{1-x^2}}{1} \frac{\text{Adj}}{\text{hyp}}$$

Now take the derivative
of $\sin y = x$.

$$\cos y \cdot y' = 1$$

$$y' = \frac{1}{\cos y}$$

$$y' = \frac{1}{\sqrt{1-x^2}}$$

So, the derivative of $\sin^{-1} x$
is $\frac{1}{\sqrt{1-x^2}}$

Inverse Trig Derivatives

$$(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\cos^{-1} x)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(\tan^{-1} x)' = \frac{1}{1+x^2}$$

$$(\cot^{-1} x)' = \frac{-1}{1+x^2}$$

$$(\sec^{-1} x)' = \frac{1}{x \sqrt{x^2-1}}$$

$$(\csc^{-1} x)' = \frac{-1}{x \sqrt{x^2-1}}$$

Ex ⑦ $y = \sin^{-1}(\sqrt{x})$ Find y' .

chain
Rule



$$y' = \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}}$$

$$y' = \frac{1}{2\sqrt{x}\sqrt{1-x}}$$

or $y' = \frac{1}{2\sqrt{x-x^2}}$